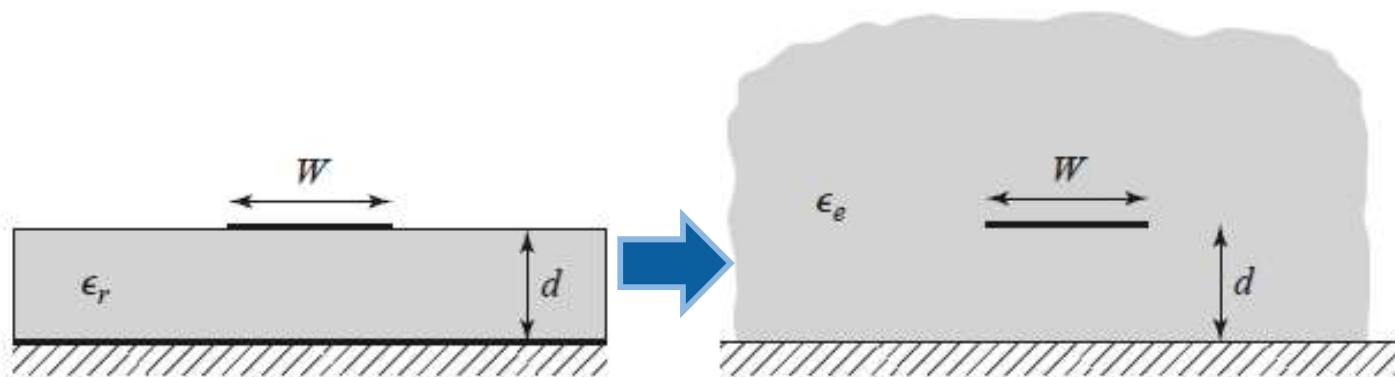


Microstrip transmission lines: ϵ_e

- Since some of the field lines are in the dielectric region and some are in air, the effective dielectric constant satisfies the relation: $1 < \epsilon_e < \epsilon_r$.
- The effective dielectric constant of a microstrip line is given approximately by: $\epsilon_e = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \frac{1}{\sqrt{1 + 12d/W}}$.



Microstrip transmission lines: $\frac{W}{d}$

- For a given characteristic impedance Z_0 and dielectric constant ϵ_r , the W/d ratio can be found as:

$$\frac{W}{d} = \begin{cases} \frac{8e^A}{e^{2A}-2} & \text{for } W/d < 2 \\ \frac{2}{\pi} \left[B - 1 - \ln(2B - 1) + \frac{\epsilon_r - 1}{2\epsilon_r} \left\{ \ln(B - 1) + 0.39 - \frac{0.61}{\epsilon_r} \right\} \right] & \text{for } W/d > 2 \end{cases}$$

$$A = \frac{Z_0}{60} \sqrt{\frac{\epsilon_r + 1}{2}} + \frac{\epsilon_r - 1}{\epsilon_r + 1} \left(0.23 + \frac{0.11}{\epsilon_r} \right)$$

$$B = \frac{377\pi}{2Z_0\sqrt{\epsilon_r}}$$

